

# 1. Project Introduction

## 1.1 Overview

This project is an end-to-end data pipeline designed to collect, process, and analyze Formula 1 telemetry data. The project focuses on providing a precise analytical comparison between the performance of drivers (such as Max Verstappen and Fernando Alonso in the 2023 Monaco Grand Prix) to understand the nuances of their driving strategies.

### Project Objectives

- Building a robust ETL pipeline to extract complex telemetry data and transform it into structured data.
  - Creating a cloud data warehouse to store data securely and scalably.
  - Designing interactive dashboards that enable decision-makers to analyze driver speeds, braking zones, and throttle usage down to fractions of a second.
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## 2. Tools and Libraries

The project relies on the latest technologies in the field of data engineering and analytics:

- **Python (FastF1):** To extract raw data from Formula 1 servers via the API.
  - **Pandas:** For transformation operations, data cleaning, and data preparation.
  - **Snowflake:** The central cloud data warehouse for storing the final tables and building data models.
  - **SQL:** To process data within Snowflake and write analytical queries.
  - **Power BI:** To connect to the data warehouse and build the final data visualization interface.
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## **3. ETL Pipeline Process**

### **3.1 Extraction Phase**

The FastF1 library was used in a Python environment to fetch the race session data.

- **Targeted Data:** Lap times and telemetry data, which includes (Speed, RPM, Throttle, Brake, Distance).

### **3.2 Transformation and Cleaning Phase**

Pandas was used to process the data before exporting it:

- Calculating the distance covered and linking it to the speed to standardize the basis of comparison between drivers.
- Handling missing values and exporting the data in CSV format.

### **3.3 Loading and Storage Phase (Snowflake)**

The table schema was created in Snowflake to upload the processed data.

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## **4. Data Modeling & SQL**

Within the Snowflake environment, SQL queries were used to prepare the data for visualization, and the operations included:

- Data aggregation to calculate the average speeds in specific sectors of the circuit.
  - Creating dimension tables if necessary (such as a table for drivers and a table for circuits).
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## **5. Data Visualization (Power BI)**

Power BI is connected to Snowflake. The dashboard contains the following charts:

- **Telemetry Line Chart:** A line chart showing distance on the horizontal axis (x) and speed on the vertical axis (y) to compare two drivers in the same lap, illustrating areas of advantage instantaneously.
- **Brake vs. Throttle Map:** An analysis of driver responses in sharp corners and a comparison of braking timing.
- **Lap Time Comparison:** Tables showing time differences in milliseconds.